

Intensive (Daily) Behavior Therapy for School Refusal: A Multiple Baseline Case Series

David F. Tolin

Sara Whiting

Nicholas Maltby

Gretchen J. Diefenbach

Mary Anne Lothstein

Surrey Hardcastle

Amy Catalano

Krista Gray

The Institute of Living, Hartford, CT

The following multiple baseline case series examines school refusal behavior in 4 male adolescents. School refusal symptom presentation was ascertained utilizing a functional analysis from the School Refusal Assessment Scale (Kearney, 2002). For the majority of cases, treatment was conducted within a 15-session intensive format over a 3-week period. Treatment elements included cognitive-behavioral therapy with the adolescent, parent training sessions, or a combination of these strategies. Treatment was effective for 3 of 4 cases in the short term. At 3-year follow-up, all 3 of the acute treatment responders had switched to alternative educational programs, although parents rated them as significantly improved and less impaired compared to pretreatment. Obstacles to treatment, and recommendations for program improvement, are discussed.

SCHOOL refusal behavior is defined as a refusal by the child or adolescent to attend school or to remain in classes for the entire school day, and may include either partial or complete school nonattendance. School refusal may be associated with signs of anxiety, depression, or disruptive behavior such as tantrums, as well as more subtle behaviors such as demure requests from a child to avoid school (Kearney, 1995). It has been estimated that approximately 5% of children engage in school refusal behavior (Granell de Aldaz, Vivas, Gelfand, & Feldman, 1984), with peak incidence during transitions between elementary, junior high, and high schools (Ollendick & Mayer, 1984).

School refusal behavior may be associated with long-term psychosocial maladjustment. Children who miss school are less likely to excel academically or fit in socially, and are at increased risk for problems of psychosocial adjustment in adulthood (Hibbett & Fogelman, 1990). Therefore, effective intervention is critical. Previous studies of cognitive-behavioral therapy (CBT) for school refusal behavior have employed a multidimensional

approach incorporating graded exposure as well as coping skills training (King et al., 1998; Last, Hansen, & Franco, 1998), parent and teacher behavior management training (King et al., 1998), or cognitive restructuring (Bernstein et al., 2000). Results of randomized controlled trials have been mixed. Two trials that emphasized imaginal and in vivo exposure and coping self-statements, with sessions directed primarily toward the child, demonstrated limited efficacy (Bernstein et al.; Last et al., 1998); in one (Bernstein et al.), CBT was not effective unless combined with imipramine. However, another trial, in which exposure and coping self-statements for the child were interspersed with parent- and teacher-directed behavior management training (King et al., 1998), resulted in significantly reduced school refusal behavior and maintained treatment gains at 3-month (King et al., 1998) and 3- to 5-year follow-ups (King et al., 2001). Of note, two of the prior studies have included few or no children with significant depression (King et al., 1998; Last et al.), despite the high rate of depression among school-refusing children (Kearney, 1993).

School refusal is a heterogeneous problem, and may serve multiple functions. Children and adolescents may engage in school refusal behavior to avoid stimuli that provoke negative affect (e.g., anxiety and depression), to

escape aversive social or evaluative situations (e.g., peer relationships, oral presentations), to receive increased attention from caregivers (e.g., disruptive behavior to stay home with one's parents), or to gain positive tangible reinforcement (e.g., finding it more rewarding to be with friends outside rather than inside school; [Kearney & Silverman, 1990, 1993](#)). Recently, researchers have paid increased attention to tailoring CBT protocols to match the specific functions of school refusal behavior (e.g., [Kearney, Pursell, & Alvarez, 2001](#)). Current best-practice guidelines recommend that interventions vary, depending on the outcome of a functional analysis, in terms of whether they are child-focused, parent-focused, or family-focused; and make differing use of strategies such as exposure, psychoeducation, cognitive restructuring, contingency contracting or other parent management strategies, or other cognitive-behavioral interventions as needed ([Kearney, 2001](#); [Kearney & Albano, 2000](#)). Similarly, based on predictor analyses, researchers have recommended tailoring CBT to address co-occurring psychopathology such as separation anxiety disorder ([Layne, Bernstein, Egan, & Kushner, 2003](#)). However, this flexibly tailored approach has received little empirical study, as previous studies ([Bernstein et al., 2000](#); [King et al., 1998](#); [Last et al., 1998](#)) have used a more fixed protocol.

The frequency of treatment sessions also merits additional consideration. Previous studies ([Bernstein et al., 2000](#); [King et al., 1998](#)) have involved 50- to 60-minute sessions, delivered 1 to 2 times per week. However, results of those trials have yielded mixed results. To the extent that exposure-based interventions are emphasized, some studies (e.g., [Foa, Jameson, Turner, & Payne, 1980](#)) have suggested that more frequent (e.g., daily) exposure sessions might be more effective than weekly treatment, although empirical results on the spacing of exposure sessions have yielded inconsistent results (see [Craske et al., 2008](#), for a review). A daily format might also yield more practical benefits, such as the ability to address the quickly compounding effects of missing school (e.g., falling behind, leading to increased distress and avoidance; [Kearney & Tillotson, 1998](#)) and the ability to correct homework problems immediately (e.g., [Tolin & Franklin, 2002](#)); indeed, daily contact with families has been recommended in cases of homework noncompliance ([Kearney, 1995](#)). This topic has received little empirical attention in school refusal. [Moffit, Chorpita, and Fernandez \(2003\)](#) conducted a single-case study involving a complex and difficult case of school refusal behavior. Although daily treatment sessions were not used, the therapist had daily phone sessions with the child and the parent, augmented by treatment sessions held at the child's school. School personnel were consulted through regular phone contact. School attendance improved, although continued difficulties were noted.

Several negative prognostic indicators have been suggested based on clinical observation ([Kearney, 1995](#)) or controlled research ([Layne et al., 2003](#)). These include poor attendance or hostility from one or more family members, noncompliance with assigned procedures, child and/or parent depression, marital conflict, long-term school refusal behavior, very low (or absent) school attendance, mixed functional profiles, and the presence of co-occurring anxiety disorders. Anecdotally, we have noted that the majority of children we have assessed with school-refusal behavior have presented with at least one (usually more) of these negative predictors. Such patients might be considered more severe or complex than those included in previous controlled trials, and therefore might be particularly good candidates for a tailored and intensive CBT approach.

The aim of the present study was to further assess the utility of intensive (daily) CBT for school-refusing adolescents. We sought to include youths with school refusal behavior severe enough to warrant referral by their school districts, and to test a flexible model of CBT that was tailored to the function of the school refusal behavior, as well as co-occurring psychopathology ([Kearney & Albano, 2000](#)). We used a multiple baseline design with four school-refusing adolescents to examine the efficacy of a 15-session intensive individual CBT tailored to school refusal function. The primary outcome, school attendance, was tracked on a daily basis; secondary outcomes, including functional impairment and global impressions of improvement, were assessed at pretreatment, posttreatment, and 3-year follow-up. In the following sections, we will describe each case and the presenting complaints. After discussing the assessment process, we will provide psychometric data and case conceptualizations for each patient. The specific interventions used for each patient will be discussed, followed by a description of treatment outcomes.

Method

Participants

The four children described here were sampled from seven consecutive referrals to the School Refusal Program at the Institute of Living's Anxiety Disorders Center and Grace Webb School in 2005. Of the seven referrals, one child was excluded from the program due to the presence of an autism spectrum disorder, another was excluded due to the presence of anorexia nervosa, and a third declined treatment, choosing instead to attend a therapeutic day school. The remaining four entered and completed treatment, and are described below.

Case 1: Chris. Chris¹ was a 16-year-old Caucasian male referred for treatment after experiencing difficulty with school attendance for approximately 1 year, with com-

¹ All names and other identifying information have been altered.

plete school absence over the past 3 to 4 months. Chris's difficulties with school attendance were associated with two main stressors. First, Chris found it difficult to get up in the morning for school because he frequently did not go to bed until 4:00 a.m. Second, Chris missed school often because of severe allergies. After missing several months of school, Chris became increasingly apprehensive about going back to school because of his concern about what his peers might think of his long absence. Chris was concerned that he would not have an adequate explanation for the amount of time he missed. He was also worried that his peers would not like him.

Case 2: Jason. Jason was a 13-year-old Caucasian male who, at the time of referral, had not attended school regularly for approximately 7 months. For Jason, school nonattendance was related to a concern over how his food allergies or asthma might affect his ability to function at school (e.g., interact with his peers during lunch, receive appropriate medical attention). When Jason became nervous at school, he would either ask to go to the nurse's office, or request that his mother pick him up from school because he was feeling nauseated. Jason's mother reported that her son was uncomfortable at school because he was less physically mature than his peers. She added that when her son was not in school, he did not experience any physical symptoms.

Case 3: Trevor. Trevor was a 15-year-old Hispanic male who was absent from school for a substantial amount of time during the 9th grade due to an illness. When Trevor attempted to return to school following his illness, he panicked in anticipation of attending school. His fears became unmanageable and incapacitating when dressing for school, riding in the car toward school, and standing at the school door. As a result of substantial absences, Trevor had to repeat the 9th grade. His parents agreed that as an alternative to the regular school program, Trevor would receive tutoring in the resource room at school. Trevor continued to experience panic even when presented with this alternative. As a result, his attendance for tutoring at the resource room was sporadic.

Case 4: Billy. Billy was a 15-year-old Caucasian male who, at the time of referral, had not attended school for approximately 3 to 4 months. Billy's school attendance difficulties began when he transitioned from a small middle school to a large high school, where he felt pressured by the academics and as if he did not fit into the social climate of the high school. After missing 1 day of school per week for approximately a month, Billy changed high schools. The school change, however, did not make him feel any more comfortable. Instead, Billy reported that when he attended school, he experienced heart palpitations, shortness of breath, and lightheadedness for the entire time he was in school. Billy began to stay home to prevent these anxiety symptoms from

returning. Billy's mother found it difficult to refuse her son's requests to miss school and ultimately quit her job to stay home with her son.

Multidisciplinary Assessment

Educational consultation. All children were referred to the program by public school officials. Following an initial telephone consultation with the school administrator, an official referral was made by sending a copy of the child's school records along with any other pertinent information. The packet generally contained background information documenting the school's attempts to remedy the child's school refusal problems, and sometimes contained grades or test scores. A clinical psychologist specializing in educational programs for emotionally disturbed children, who was a member of the treatment team, reviewed the packet and made an initial determination about whether the School Refusal Program would be appropriate. This psychologist also consulted by telephone with school staff members such as the school counselor.

Clinical evaluation. Children were then examined by a clinical psychologist with expertise in anxiety-related disorders. The psychologist consulted with the educational specialist described in the previous section, and then administered a structured interview, child and parent self-report measures, and a clinician rating. The Anxiety Disorders Interview Schedule–Child Version (ADIS-C; [Albano & Silverman, 1996](#)) was used to obtain diagnostic information and a description of school-related behaviors. The ADIS-C is a structured interview of *DSM-IV* ([American Psychiatric Association, 1994](#)) diagnostic criteria for anxiety, mood, and other disorders. The School Refusal Assessment Scale–Revised (SRAS-R; [Kearney, 2002](#)) was used to identify the specific function of school refusal behavior from the child's and parent's perspective. The SRAS-R is a 16-item self-report measure that assesses four possible reasons for school refusal: avoidance of negative affect, escape from social evaluation, attention-getting behavior, and positive tangible reinforcement. The highest-rated scale is presumed to reflect the primary function of school refusal behavior, with scores within 0.5 points of each other taken to reflect mixed functional profiles ([Kearney & Silverman, 1999](#)). The SRAS-R shows acceptable test-retest reliability, and functional scores show an expected pattern of correlations with measures of internalizing and externalizing disorders; however, parent-child agreement is modest ([Kearney, 2002](#)). Other self-report measures included the Children's Depression Inventory (CDI; [Kovacs, 1985](#)), a 27-item self-report measure given to children to identify symptoms of depression experienced over the past two weeks, and the Multidimensional Anxiety Scale for Children (MASC; [March, Parker, Sullivan, Stallings, & Conners, 1997](#)), a 39-item self-report measure used to

assess anxiety in children by examining physical symptoms, harm/avoidance, social anxiety, and separation/panic. Global severity of illness was ascertained utilizing the Clinician's Global Impression (CGI; Guy, 1976). The CGI consists of a 7-point rating scale of severity from "normal, not at all ill" to "extremely ill," as well as a 7-point scale of improvement from "very much worse" to "very much improved." Additionally, parents completed a parent-report version of the CGI. Parent reports of children's psychosocial impairment in schooling, social, and family domains was assessed using a modified version (Whiting & Tolin, 2008) of the Sheehan Disability Scale (SDS; Leon, Shear, Portera, & Klerman, 1992).

Finally, the child and parents were given a binder of School Attendance Logs, developed by the authors, in which they recorded the child's daily school attendance. These logs were reviewed daily and used to document progress toward school attendance. Because of the naturally occurring variability in therapist availability, scheduling, etc., the daily ratings could be subject to multiple baseline analysis, in which the duration of pretreatment is allowed to vary across individuals. Multiple baseline analyses are particularly helpful in differentiating the specific effects of treatment from certain nonspecific effects such as time and regression to the mean (Barlow & Hersen, 1984). After the completion of 15 therapy sessions (described later), the child and parents completed the measures again and were interviewed about their progress and ongoing needs. A 3-year telephone follow-up assessment was performed with parents, including the parent-rated CGI, modified SDS, and a brief, semistructured interview about their children's progress and educational outcomes.

Assessment Results

Pretreatment data for all four children are depicted in Table 1.

Case 1: Chris. According to the ADIS-C, Chris met diagnostic criteria for Major Depressive Disorder, Single Episode, Moderate; and Social Phobia. On the SRAS-R, Chris's mother reported a mixed functional profile, in which his school refusal behavior was associated with efforts to avoid experiences of negative affect and escape negative social evaluation. However, Chris described a singular functional profile, in which he highlighted the role of positive tangible reinforcement. Other self-report measures suggested high levels of depression and anxiety (particularly physical symptoms and social anxiety).

Case 2: Jason. Jason met diagnostic criteria for Specific Phobia (situational type). On the SRAS-R, Jason and his mother both identified a singular function of school refusal behavior; however, they differed in terms of the identified function. Jason identified avoidance of negative

Table 1
Pretreatment Data

	Chris	Jason	Trevor	Billy
SRAS-R (Child version)				
Avoidance of Negative Affect	1.50	4.17	3.83	2.50
Escape from Social Evaluation	1.67	1.00	1.17	0.17
Attention-Getting Behavior	0.00	0.83	2.00	2.17
Positive Tangible Reinforcement	4.00	1.17	2.00	2.67
SRAS-R (Parent version)				
Avoidance of Negative Affect	3.17	3.83	4.17	6.00
Escape from Social Evaluation	3.50	3.50	3.17	5.66
Attention-Getting Behavior	0.67	4.33	0.50	4.40
Positive Tangible Reinforcement	0.67	3.00	1.33	0.50
CDI	23	10	9	12
MASC				
Physical Symptoms	26	23	16	2
Harm Avoidance	10	11	17	8
Social Anxiety	23	6	11	3
Separation/Panic	4	7	4	5
Total	63	47	48	18
CGI-S (Therapist)	6	6	6	6
CGI-S (Parent)	5	7	6	7
SDS-Parent				
Schooling	9	10	9	10
Social	8	10	6	9
Family Life	9	10	5	9
Total	8.7	10.0	6.7	9.3

Note. SRAS-R=School Refusal Assessment Scale-Revised; CDI=Child Depression Inventory; MASC=Multidimensional Anxiety Scale for Children; CGI-S=Clinical Global Impression-Severity (1=normal, not at all ill; 2=borderline mentally ill; 3=mildly ill; 4=moderately ill; 5=markedly ill; 6=severely ill; 7=extremely ill); CGI-I=Clinical Global Impression-Improvement scale (1=very much improved; 2=much improved; 3=minimally improved; 4=no change; 5=minimally worse; 6=much worse; 7=very much worse); SDS=Sheehan Disability Scale (0=not at all; 1-3=mild; 4-6=moderate; 7-9=marked; 10=very severe).

affect, whereas his mother identified attention-getting behavior. The CDI and MASC were consistent with elevated depression and physical symptoms of anxiety.

Case 3: Trevor. Trevor met diagnostic criteria for Specific Phobia (situational type). SRAS-R scores for Trevor and his mother both identified a primary function of avoiding negative affect. Although Trevor did not appear depressed on the CDI, he reported physical symptoms of anxiety on the MASC.

Case 4: Billy. Billy was diagnosed with Specific Phobia (situational type) and Depressive Disorder Not Otherwise Specified. On the SRAS-R, Billy and his mother both identified a mixed functional profile; however, the profiles differed. Billy identified positive tangible reinforcement and avoidance of negative affect as the primary functions of his school refusal behavior, whereas his mother identified avoidance of negative affect, escape

from social evaluation, and attention-getting behavior. Billy reported depressive symptoms on the CDI, but generally denied symptoms of anxiety on the MASC.

Treatment

General procedures. Following the initial evaluation, the treatment team met with the child and parent(s) for feedback and recommendations. Different treatment options, including the School Refusal Program, were presented. The treatment program entailed 15 sessions of CBT, delivered 5 days per week over 3 weeks (although, as described below, Billy's 15 sessions were administered over 8 weeks due to a variety of factors). Sessions lasted approximately 90 to 120 minutes. Treatment was adapted from comprehensive guidelines published by Kearney (2001) and Kearney and Albano (2000). Following the examples from those programs, treatment was tailored according to a functional analysis, based in part on the SRAS-R.

Specific CBT interventions were derived from established school refusal programs (Kearney, 2001; Kearney & Albano, 2000) as well as from other child anxiety interventions (Kendall & Hedtke, 2006; King, Heyne, & Ollendick, 2005; Roblek & Piacentini, 2005). Direct, graded exposure to feared situations (which could include in vivo exposure to school-related stimuli, in vivo exposure to social interactions, or interoceptive exposure to feared internal sensations, depending on the case conceptualization) was a central feature of treatment for all children, as was parent training in contingency management. Other interventions were used as needed, based on the case conceptualization. These could include cognitive restructuring, behavior rehearsal (skill training, role-playing), relaxation training, motivational interviewing, and environmental modifications (including family counseling to alter maladaptive interactional patterns). Flexibility in treatment planning, rather than rigid adherence to a fixed protocol, was emphasized (see Kendall, Chu, Gifford, Hayes, & Nauta, 1998), and clinicians were free to select specific CBT interventions as needed. All treatments included the child and parent(s) in sessions, although, depending on the case, more or less parent-directed sessions could be used (and, at times, parent sessions were held in the absence of the child). Flexibility was emphasized, in which sessions could take place in the therapist's office, at the child's home, or at the school. During treatment, the educational specialist on the treatment team communicated directly with the therapist and reported to the referring school district on the child's progress.

Case 1: Chris. As noted above, Chris's assessment suggested a complicated functional profile, with possible contributors including avoidance of negative affect, escape from social evaluation, and positive tangible

reinforcement, as well as substantial depression and social anxiety. Specific interventions included:

- *Social problem solving and behavior rehearsal.* In the initial session, Chris discussed his apprehension about returning to school, stating he did not know how to explain his prolonged absence to his peers. It was suggested that he either brush off any such questions, or have a "cover" story, such as being home-schooled. The therapist role-played such interactions with Chris.
- *Sleep hygiene.* Chris described difficulty getting out of bed in the morning to go to school; the therapist learned that he stayed up late at night watching TV and playing video games, and therefore was fatigued throughout the day. Chris was instructed to stop taking naps during the day, in order to help regulate his sleep/wake cycle. This proved helpful, and within 24 to 48 hours Chris was able to sleep at night and remain awake through the day.
- *Contingency management.* The therapist recommended that Chris's parents restrict his use of the TV and game system, and make their use contingent on adherence to the targets of treatment. Chris's father, an electrician, was able to switch off the power to the TV and game system, and turn it on only when Chris had met his daily goals. Chris's use of the family car was similarly made contingent on school attendance.
- *Graded exposure to school.* At the fourth treatment session, the therapist accompanied Chris to school, where Chris went on to attend most of the day. Over subsequent days, Chris was encouraged to attend for progressively longer periods of time. Midway through treatment, the therapist also attended a parent-teacher conference at the school to discuss ways that the school could help Chris maintain his gains.
- *Family counseling.* Initially, Chris's father was highly critical and rejecting of Chris. Chris's mother, on the other hand, tended to nag Chris repeatedly, causing resentment. The therapist spoke with the parents, without Chris present, to discuss more effective ways to manage the problem behaviors. Chris's father became much more supportive of Chris, and started spending more enjoyable time with him. Chris's mother was able to "back off" and allow Chris more flexibility in his decisions.

Case 2: Jason. Jason's treatment focused on school refusal behavior that was due to his excessive fear of having an asthma or allergy attack at school. Specific treatment components included:

- *Graded exposure to school.* At the first session, the therapist instructed Jason and his parents to establish a morning routine in which Jason would get up in time for school. He and his parents would then drive by the

school, regardless of whether Jason actually intended to attend school that day. Two days later, he was instructed to enter the school to visit his guidance counselor, although class attendance was not required. When Jason did not follow through with this latter assignment, the therapist immediately scheduled an out-of-office visit in which he and Jason went to the school building together. This session went well, and by the next day (Session 5), Jason was able to spend 2 hours doing classwork in the school library. In subsequent visits Jason was able to spend progressively longer periods of time in the school building, although he was noted to continue to complain of anxiety and request to leave school. The therapist made repeated morning visits to Jason's house in order to encourage him to continue attending.

- *Interoceptive exposure to signs of physiological arousal.* Because Jason appeared to misinterpret benign physiological arousal sensations as signs that he was about to have an asthma or allergy attack, the therapist instructed Jason to engage in interoceptive exposures in the office such as hyperventilating and running up and down stairs. The exercises elicited feelings of anxiety and worry that he would have an asthma attack; these fears decreased over the course of the session. These exercises were done after consultation with Jason's pulmonologist. To help address Jason's fears of an asthma attack, the pulmonologist provided Jason with a peak flow meter so that he could determine objectively whether he was indeed having an asthma attack at school.
- *Contingency management.* At the initial session it was agreed that Jason would not be able to use the television or computer during school hours. Access to the computer was blocked by putting a password on the computer so that only Jason's parents could authorize its use. Time using the TV and computer were used as reinforcers for successful homework completion. Between Sessions 7 and 8, Jason's mother called the therapist to report that he had left school grounds and was walking home (Jason had informed her of this via cell phone). The therapist instructed Jason's mother to call the police to pick him up and return him to school. When Jason's mother called him to inform him that she intended to do so, Jason returned to school on his own and remained there for the assigned amount of time. At the next session, Jason appeared increasingly motivated to participate in plans to return to school full time.

Case 3: Trevor. Trevor's school refusal behavior was related to his fear of having panic attacks at school. Specific treatment components included:

- *Graded exposure to school.* At the second treatment session, the therapist accompanied Trevor and his

mother to the school building. It was noted that Trevor's anxiety increased as they drove toward the school, and appeared to peak immediately upon exiting the car in the parking lot. Trevor therefore practiced walking away from the car repeatedly, while imagining that he was going to spend the entire day at school. His self-reported fear diminished with repeated practice, and Trevor was instructed to continue this exercise until the next session, which he did. The therapist accompanied Trevor to school again at Session 4, and Trevor was able to spend the day at school in the resource room, reading with a tutor. By Session 5, Trevor was being tutored in an empty classroom rather than the resource room. He attended his scheduled classes by Session 6. By Session 8, exposure to riding the school bus home was also incorporated.

- *Contingency management.* At Session 7, Trevor's mother reported that he had experienced a panic attack in the car on the way to school that day, did not enter the school building until 10:00 a.m., and had requested to leave early (with support from school staff, he nevertheless remained in school). The therapist and Trevor's mother agreed that if Trevor exhibited school refusal behavior in the car, she would park the car at school, take the keys, and walk home. During one school visit, the therapist modeled this strategy for Trevor's mother. Use of the home computer was also made contingent upon meeting daily attendance targets.
- *Modification of school schedule.* Trevor expressed unhappiness with his school schedule. Specifically, his scheduled time in the resource room led him to feel stigmatized as having a "learning disability." The therapist met with school administrators and it was agreed that Trevor would be able to attend sophomore study hall instead of the resource room.

Case 4: Billy. For Billy, treatment focused on increasing motivation for attending school and reducing social anxiety related to school attendance. Treatment was not administered daily due to several factors including snow day cancellations, vacations, and Billy's refusal to attend sessions. Specific treatment components included:

- *Motivational interviewing.* Billy presented with significant resistance toward the goal of reducing his school refusal behavior. Given a history of oppositional behavior with parents and previous therapists, a collaborative, motivational interviewing approach was used to engage Billy. This process was initiated in the first session and continued throughout treatment to greater or lesser extents as Billy's level of motivation and self-efficacy for change waxed and waned.

- *Establishing a routine.* Billy's routine prior to beginning treatment lacked structure (e.g., he would sleep until noon when he was awoken by his mother to attend tutoring at the local library). During the second session a new morning routine was established, which included driving to the school at the same time every day as if he were attending school.
- *Contingency management.* Contingency management was discussed but this technique did not become a component of treatment, given that neither Billy nor his mother could identify external reinforcers. Discussions of implementation of this technique also prompted power struggles between Billy and his mother.
- *Cognitive therapy and relaxation strategies.* Both cognitive therapy and relaxation strategies were introduced during the second session with the goal of providing Billy with anxiety management skills that would allow him to engage in exposure therapy. While Billy's SRAS-R profile might indicate that social anxiety was not a significant component to his school refusal, his therapist disagreed with these data based on Billy's psychosocial history and discussions with his mother and school personnel. Cognitive therapy was emphasized to help Billy identify fears underlying his school refusal behavior. This process was complicated by some narcissistic personality traits wherein Billy expressed feeling "superior" to his peers in school, thus he was reluctant to admit that his anxiety stemmed from how his peers would perceive him. Over time, Billy began to acknowledge a substantial social anxiety component to his school refusal behavior. He identified fears of making mistakes, forgetting homework, being embarrassed, getting lost, and going to the wrong classroom. He also identified significant perfectionism and black-and-white thinking associated with school work. The therapist employed Socratic questioning, examination of alternative explanations, and behavioral experiments to counter Billy's maladaptive beliefs. The therapist also engaged Billy in social skills training and role-play exercises to address one of his primary social fears—answering questions from fellow students about why he has been absent from school.
- *Graded exposure to school.* Following cognitive therapy, Billy actively engaged in developing an exposure hierarchy and identifying adaptations that would assist him with the transition (e.g., being allowed to eat lunch in the library). The next eight sessions focused on imaginal and in vivo exposures. Imaginal exposures were completed without difficulty. However, beginning in vivo exposure was challenging; Billy reported significant anxiety anticipating walking into the school. To gradate this exposure, his first in vivo school exposure was done on a Saturday. Despite the

modification, Billy continued to report extreme anxiety. He initially refused to complete the exposure, but with encouragement he did participate in driving to the school and sitting in the school parking lot. Billy refused to leave the car even after altering the exposure to gradate the experience further to walking around the parking lot instead of entering the school. During this session it became clear that Billy's extreme anxiety and intolerability of distress were additional challenges to completing exposures and reducing his school refusal. After this experience, Billy's motivation began to wane and he began to question the approach. Due to this reaction, further sessions were put on hold for a brief time so that in vivo exposures could be performed during a regularly scheduled school break. Billy and his mother were instructed to continue driving to the school and walking the grounds daily to facilitate the next in vivo exposure therapy session. These assignments were not completed. In therapy sessions with in vivo exposures during the school break, Billy was able to walk into the school, meet with school personnel, and tour the location of where he would attend tutoring and classes. In the final sessions of the program, Billy eventually did attend tutoring in the school; however, the consistency of his attendance was variable due to Billy refusing to attend sessions, scheduling conflicts by the tutor, Billy's asthma attacks, and family issues contributing to significant distress.

- *Family counseling.* While Billy's mother was actively involved in treatment, at times she failed to consistently implement the program. During exposures, the therapist noted Billy's mother's behavior to be unproductive and unsupportive. In addition, Billy and his mother reported increased parent-child conflict following the exposures. To address this conflict, parent training was conducted with Billy's mother and communication/conflict resolution skills were introduced to Billy and his mother. There were additional complex, antagonistic family dynamics which limited the involvement of other caregivers (e.g., father, stepfather) in his treatment. Given the time-limited and focused nature of the school refusal program, these family issues were not worked through during this treatment program. Instead, a referral was made for family therapy at the conclusion of the program.

Treatment Results

Figure 1 depicts school attendance for the four children over time, verified by parents' logs. The *C* statistic (Young, 1941) was used to determine nonrandom changes in school attendance over repeated measurements during the baseline and treatment phases. In all cases, no significant change over time was observed during the baseline phase (Chris $C=0.00$, Jason $C=0.00$,

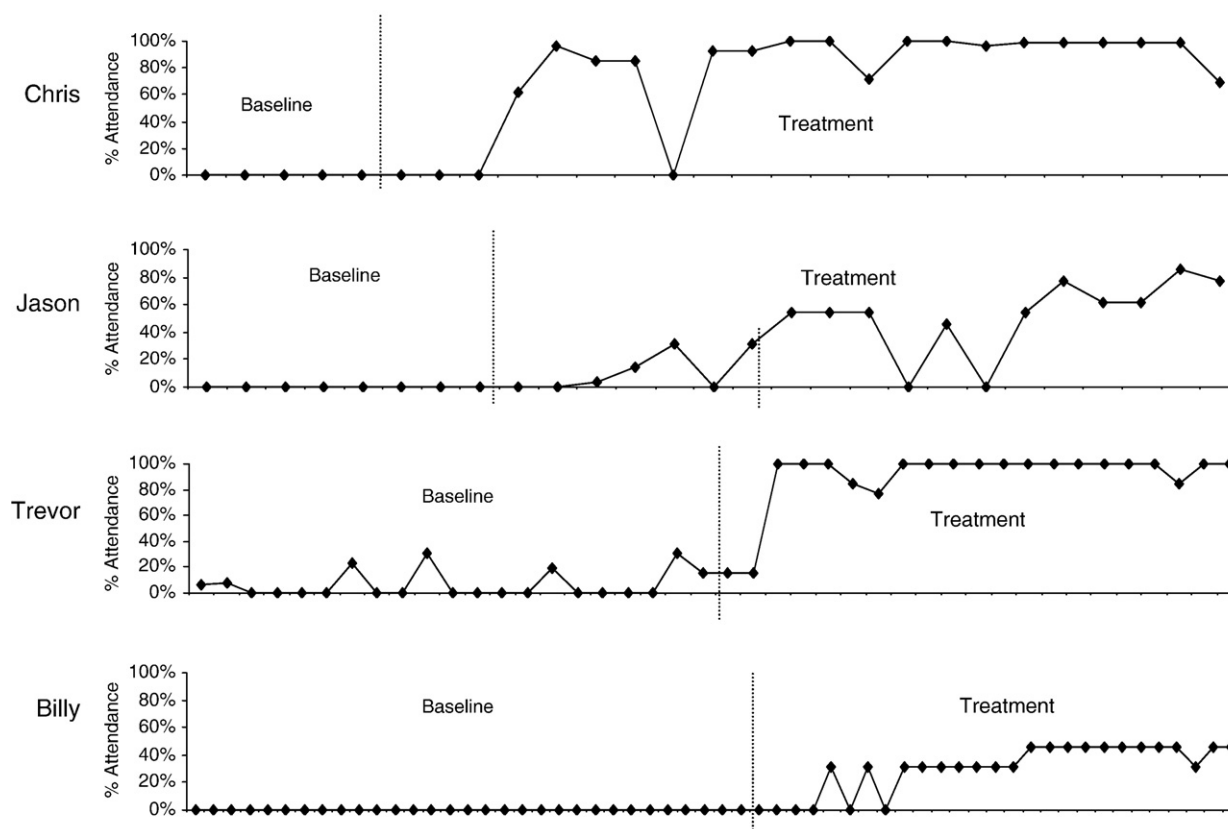


Figure 1. Daily School Attendance for Four School-Refusing Children Treated with Intensive (Daily) Behavior Therapy.

Trevor $C=0.24$, Billy $C=0.00$, all $p's > .05$). When the intensive CBT program was initiated, however, all four children showed an improvement in school attendance behavior (Chris $C=0.60$, $p=.001$; Jason $C=0.56$, $p=.004$; Trevor $C=0.67$, $p=.001$; Billy $C=0.67$, $p=.001$). All improvement during the treatment phase represented a significant improvement in trend over the baseline phase (Chris $C=0.78$, $p=.001$; Jason $C=0.67$, $p=.001$; Trevor $C=0.94$, $p=.001$; Billy $C=0.85$, $p=.001$).

Case 1: Chris

After receiving 3 days of treatment, Chris was consistently attending school part of the day. At the time of discharge, Chris had attended school every day for 2 consecutive weeks. Chris had been 5 to 10 minutes late regularly for homeroom but was attending all of his academic classes. Recommendations for maintaining success included having Chris's parents continue to establish contingency plans. For example, Chris's parents were planning to buy a car for their son; as a result, it was suggested that car use be dependent upon attending school. Follow-up sessions were also scheduled for Chris. Recommendations to the school included exempting Chris from episodic school tardiness. However, if a pattern of school nonattendance occurred, the school

was instructed to call Chris's parents, to avoid having the symptoms develop further.

Chris's school attendance improved from no attendance at all to nearly 100% attendance, although a slight drop was noted immediately prior to termination of treatment. Pretreatment, posttreatment, and follow-up data are listed in Table 2. Examining the highest-rated SRAS-R score (reflecting the main function of school refusal behavior), Chris's parents reported decreased efforts to avoid negative affect and escape social evaluation. Chris did not complete the posttreatment measures. Both the therapist and Chris's mother rated his condition as "borderline ill" and "very much improved." On the parent-rated SDS, Chris's schooling and family life impairment decreased from marked to moderate and his social impairment decreased from marked to not at all impaired.

At the 3-year telephone follow-up, Chris's mother reported that he had stopped attending school shortly after completing treatment, worked for 2 1/2 years, and then enrolled in an alternative night-school program, 4 days per week, to earn his high school diploma. At the time of follow-up, Chris had been attending this program with close to 100% attendance for 2 months. His mother also reported that Chris had plans to complete his high school diploma and attend a 2-year college. His condition

Table 2
Changes in Self-Report Measures from Pre- to Posttreatment and 3-Year Follow-up

	Chris			Jason			Trevor			Billy		
	Pre	Post	FU	Pre	Post	FU	Pre	Post	FU	Pre	Post	FU
SRAS-R (Child version)												
Avoidance of Negative Affect	1.50	–	–	4.17	2.17	–	3.83	–	–	2.50	2.83	–
Escape from Social Evaluation	1.67	–	–	1.00	0.67	–	1.17	–	–	0.17	3.00	–
Attention-Getting Behavior	0.00	–	–	0.83	0.50	–	2.00	–	–	2.17	2.50	–
Positive Tangible Reinforcement	4.00	–	–	1.17	3.17	–	2.00	–	–	2.67	3.00	–
SRAS-R (Parent version)												
Avoidance of Negative Affect	3.17	1.50	–	3.83	4.83	–	4.17	–	–	6.0	4.33	–
Escape from Social Evaluation	3.50	1.67	–	3.50	3.00	–	3.17	–	–	5.66	5.00	–
Attention-Getting Behavior	0.67	0.00	–	4.33	1.83	–	0.50	–	–	4.40	3.83	–
Positive Tangible Reinforcement	0.67	1.17	–	3.00	2.67	–	1.33	–	–	0.50	1.17	–
CGI-S (Therapist)	6	2	–	6	4	–	6	2	–	6	5	–
CGI-I (Therapist)	–	1	–	–	2	–	–	1	–	–	3	–
CGI-S (Parent)	5	2	3	7	5	4	6	–	1	7	3	–
CGI-I (Parent)	–	1	1	–	2	2	–	–	1	–	2	–
SDS (Parent)												
Schooling	9	5	6	10	7	5	9	–	0	10	4	–
Social	8	0	5	10	2	8	6	–	0	9	7	–
Family Life	9	4	6	10	5	4	5	–	0	9	7	–
Total	8.7	3.0	5.7	10.0	4.7	5.7	6.7	–	0.0	9.3	6.0	–

Note. SRAS-R=School Refusal Assessment Scale-Revised; CGI-S=Clinical Global Impression-Severity (1 = normal, not at all ill; 2 = borderline mentally ill; 3 = mildly ill; 4 = moderately ill; 5 = markedly ill; 6 = severely ill; 7 = extremely ill); CGI-I = Clinical Global Impression-Improvement scale (1 = very much improved; 2 = much improved; 3 = minimally improved; 4 = no change; 5 = minimally worse; 6 = much worse; 7 = very much worse); SDS=Sheehan Disability Scale (0 = not at all; 1-3 = mild; 4-6 = moderate; 7-9 = marked; 10 = very severe).

was rated "mildly ill" and "much improved," although he was also described as exhibiting moderate impairment in schooling, social functioning, and family life.

Case 2: Jason

Initially, Jason appeared to be only minimally invested in treatment and required pressure to engage in school or anti-anxiety related behaviors. Jason's lack of motivation was based upon his belief that anxiety symptoms represented the onset of a life threatening medical condition. By the end of treatment, however, Jason was better able to discriminate between anxiety and the onset of asthma. Overall, Jason exhibited a moderate response to treatment. By the end of treatment, he was attending school for part of the day. At the time of discharge, Jason was attending three to four classes per day and remaining in school for 4.5 to 5.0 hours. Recommendations at the time of discharge included continued school attendance with a goal of attending all classes within a 2-week time period. Parents were also instructed to continue with both the contingency management and interoceptive exposure, such as running stairs, which increased Jason's ability to tolerate anxiety and promoted habituation. The school nurse also played a crucial role in removing the opportunity for Jason to avoid class.

At the end of treatment, Jason's school attendance improved from none at all to nearly 80% attendance.

According to Jason's self-report on the SRAS-R, his avoidance of negative affect (the highest-rated scale) decreased following treatment; although his mother had identified attention-getting behavior as the primary function, this score decreased as well. The therapist and Jason's mother rated him as "much improved"; the therapist rated his overall posttreatment severity as "moderately ill," whereas his mother rated him as "markedly ill." Jason's mother reported decreased impairment in schooling (very severe to marked), social functioning (very severe to mild), and family life (very severe to moderate).

At a 3-year follow-up, Jason's father reported that, following treatment, Jason attended multiple private high schools and that he continued to have attendance difficulty at each school. He also reported multiple periods of long-duration school nonattendance. At the time of follow-up, Jason had recently started attending a therapeutic school with an 80% attendance rate over 2 weeks. Jason's father continued to rate him as "moderately ill" and "much improved," with decreased (now moderate) schooling impairment but increased (now marked) social impairment.

Case 3: Trevor

By the fourth session, Trevor was staying in school for the academic day. By the sixth session, he was attending all of his classes. Early in treatment, Trevor had two anxiety

episodes. While Trevor missed class as a result of these anxiety episodes, he remained in school. This experience gave him confidence to stay in school despite feeling anxious. Trevor only had one other anxiety episode during treatment and was able to stay in school and return to class quickly on this occasion. At discharge, Trevor was instructed to continue individual treatment at the Anxiety Disorders Center to monitor progress and provide a resource for quick intervention. In addition, a relapse prevention plan was put into place. This plan included remaining at school even when anxious. If Trevor was unable to leave the car when at school, he was instructed to stay in the car in the school parking lot until anxiety passed. He was also instructed to use the resource room or special services if he felt as if he could not stay in the classroom. Trevor was told that he should return to class as soon as possible. He was instructed to see his therapist if he had difficulty following the plan.

Trevor's attendance improved from 0%–20% at baseline to 100%. Unfortunately, neither Trevor nor his parents completed self-report measures at posttreatment; however, the therapist rated him as "borderline ill" and "very much improved."

At the 3-year follow-up, Trevor's mother reported that, following treatment, Trevor completed the ninth grade but due to absences that occurred prior to treatment, he was asked to repeat the grade for the third time. Rather than repeat the grade, Trevor chose to take, and subsequently passed, the high school general equivalency exam. His mother reported that at the time of follow-up, Trevor was attending a 2-year college with the hopes of transferring to a 4-year college to earn a bachelor's degree. His mother rated him as "not at all ill" and "very much improved." Similarly, Trevor's mother reported no current impairment in schooling, social or family life.

Case 4: Billy

At the beginning of treatment, Billy was not attending any tutoring or classes within school. By the end of treatment, he was consistently attending 2 hours of tutoring and one class at school per day, but was unwilling to increase his attendance further. At the time of discharge, Billy was instructed to continue to attend tutoring and one class for the remainder of the school year. Billy was discharged from the school refusal program very close to the end of the school year. In addition to recommendations for Billy to continue practicing the skills learned during the school refusal program, his discharge plan included recommendations to continue outpatient psychotherapy and pharmacotherapy, to begin family therapy, and to continue tutoring at the school over the summer. To prepare Billy for the transition to school in the fall, three follow-up sessions were planned. At the

first follow-up meeting, the therapist was informed that the family had not followed the recommendations provided at discharge, primarily because Billy was "doing so well" over the summer that the family believed these services were not needed. Billy presented at the follow-up meeting with a high degree of confidence and easily entered the school and walked the halls. He stated that he did not anticipate problems with attending once school began. He subsequently refused to attend the last two follow-up meetings.

Of the four children, Billy showed the least improvement in school attendance. By the end of treatment, he attended school approximately 40% of the time; his mother recontacted the therapist several months later to report that he did not return to school in the fall and that she was looking into a private school placement for him. Billy and his mother's SRAS-R showed minimal decreases or increases in all of the assessed functions of school refusal behavior from pre- to posttreatment. The therapist rated him as "markedly ill" and "minimally improved," whereas his mother rated him as "mildly ill" and "much improved," with reduced (now moderate) schooling impairment but minimal change (marked) in social and family life impairment. A 3-year follow-up could not be obtained for Billy.

Discussion

As has been the case with previous controlled trials (Bernstein et al., 2000; King et al., 1998; Last et al., 1998), the present results provide a mixed picture of treatment efficacy for CBT with school-refusing youth. On one hand, three out of the four children participating in this program showed demonstrable and meaningful improvement in school attendance and were able to re-enter the public school system. At posttreatment, average therapist and parent CGI-S ratings decreased from 6 (severe) to 3 (mild), and parent CGI-S ratings at follow-up remained at an average of 3 (mild). Average functional impairment, based on total parent-rated SDS scores, decreased from 9 (marked) to 5 (moderate) at posttreatment, and 4 (moderate) at follow-up.

On the other hand, none of the children sustained 100% attendance over a long period of time, and all three of the families we were able to contact at follow-up reported that their child had eventually opted for an alternative educational plan (night school for Chris, therapeutic school for Jason, general equivalency exam for Trevor). Assuming no further improvement in Billy (who was lost to follow-up), we obtained a 0% long-term success rate if "success" is defined as graduation from the baseline educational program. It could certainly be argued, however, that these educational adjustments reflect a positive change and that CBT played a key role in facilitating some kind of ongoing education; parents of

all three youths who could be reached at follow-up described their child's condition as "much improved" or "very much improved," with decreased impairment ratings from baseline.

Unlike previous controlled trials (Bernstein et al., 2000; Last et al., 1998), we employed a flexible, individually tailored approach based on a functional analysis of each child's school refusal behavior (Kearney, 2001). Depending on the clinical presentation, therapists chose whether to meet with the child, parents, or both. Sessions could take place in the office, in the home, or at the school. Specific interventions could include exposure, cognitive restructuring, skill building, role-playing, or parent management training. One unexpected difficulty with implementing a functionally tailored treatment was disagreement between the child and parent about the specific function of the school refusal behavior. As described above, three out of the four children in this program disagreed with their parents about the primary function(s) of the behavior. This finding is consistent with previous research showing limited parent-child agreement on the SRAS-R (Higa, Daleiden, & Chorpita, 2002; Kearney, 2002). This does not necessarily constitute a problem with the SRAS-R itself, as children and parents may indeed have different ideas about the causes of school refusal behavior. Clinically, therapists took both perspectives into account, and reconciled them using clinical interviews and their own behavioral observations. Indeed, disagreements between children and parents about why the child was refusing school often proved to be a useful starting point for clinical discussions.

Although the intensive (daily) CBT format does not approximate the traditional notion of massed exposure (Cain, Blouin, & Barad, 2003), meeting daily may offer several potential advantages over more traditional, weekly sessions (for a discussion, see Tolin & Franklin, 2002). Because this CBT program involves a great deal of homework (self-monitoring, exposure, parent management, etc.), meeting with the child and parents within 24 hours of each homework assignment may allow for the quick remediation of problems in homework implementation, rather than forcing the child to wait a week (or longer) before such problems can be corrected. However, this intensive form of treatment delivery also has its drawbacks. Frequent visits are often impossible for children or their therapists; thus, availability is decreased. Intensive programs are more costly, at least in the short term, and third-party payers are often reluctant to reimburse for such programs. It is potentially noteworthy that all four of the children described in this report presented with one or more of the negative prognostic indicators described by previous authors (Kearney, 1995; Layne et al., 2003). For example, both Chris and Billy

were diagnosed with a depressive disorder; Jason also exhibited elevated depression scores. All four presented with mixed functional profiles, and all four had a lengthy history of very low or no school attendance. Marked family conflict was noted for Chris and Billy. These features might argue for the necessity of a more intensive treatment approach than has been used in previous controlled studies, although additional research is clearly needed in order to determine optimal patient-treatment matching.

A novel, and potentially useful, aspect of the present program was a multidisciplinary approach, in which educational and clinical specialists worked as a team. While the clinical psychologist provided direct treatment to the child and parents, the educational specialist provided initial recommendations for modifications within the school, and served as a liaison with school officials. Given the complexity and difficulty inherent in treating school refusal behavior, the use of multidisciplinary teams is likely to be an important addition to usual practice (Kearney & Bates, 2005).

Billy showed the least improvement with treatment. Although Billy's school attendance did increase significantly, it did not do so sufficiently to allow him to re-enter the school system. Examination of the factors that differentiated Billy and his treatment from those of the other three children, therefore, may be of use. Billy's depression diagnosis and high CDI score do not seem explanatory, as Chris also carried a depression diagnosis and had a higher CDI score. However, certain aspects of Billy's depression may have played a role, in that it was exceptionally difficult for Billy to identify any meaningful rewards. Billy's antagonistic relationship with his mother also appeared to contribute to this problem, in that Billy seemed unwilling to suggest any reinforcers which his mother could then take from him. Therefore, contingency management procedures proved difficult for the parents to implement, and treatment focused more on individual sessions with Billy. The operant aspects of treatment might be particularly important, especially for children, such as Billy, who report at least some positive reinforcement of school refusal behavior. In these cases, parents might require additional training to simplify behavioral requests, assess whether these behavioral requests are clearly understood by the child, reward appropriately, and establish appropriate consequences for both adherence and nonadherence to established plans.

Billy's mother described a mixed functional profile on the SRAS-R, with more elevated scores than the other children; this may have also contributed by complicating the therapist's ability to tailor this relatively brief treatment (Kearney, 1995; Kearney et al., 2001). Kearney and colleagues (2001) describe the successful treatment

of two such children. Although their program was thematically similar to ours, they used fewer sessions delivered over a longer period of time. Lack of parental follow-through with homework assignments was also noteworthy in Billy's case, suggesting a need to address nonadherence immediately.

The treatment was interrupted on several occasions by weather, school and family vacations, and Billy's refusal to attend some treatment sessions. Deviations from the intensive format might have been detrimental to treatment success. One possible solution to this potential problem might include minimizing a child's breaks from school, such as finding alternative activities for the child that simulate an academic environment during school vacations. Increasing the duration of treatment and the intensity of treatment follow-ups, perhaps especially over school breaks, might also assist in sustaining successful treatment outcomes. Even for successfully treated children, the risk of future school refusal behavior remains high after vacations or school breaks. This obstacle to successful school attendance might be addressed by establishing parent support and training programs that would give parents the opportunity to express concerns, receive support, and practice adaptive strategies for managing behavior problems.

Overall, therefore, the results of the present case series are very much open to interpretation. Although the primary aim of long-term return to (and eventual graduation from) the school of origin was not met, three of four youths showed substantial short-term improvement in school attendance, and their parents reported at follow-up that they had successfully pursued alternative educational programs and were considered improved. Additional dismantling and preclinical research would be useful in determining optimal interventions for children with mixed functional profiles such as those described here. In addition, research is needed to examine the impact of intensive (daily) treatment, compared to the more traditional weekly therapy conducted in most clinics.

References

- Albano, A. M., & Silverman, W. K. (1996). *Guide to the use of the Anxiety Disorders Interview Schedule for DSM-IV: Child and parent versions*. New York: Oxford University Press.
- American Psychiatric Association. (1994). *Diagnostic and statistical manual of mental disorders*, (4th ed.) Washington, DC: Author.
- Barlow, D. H., & Hersen, M. (1984). *Single case experimental designs: Strategies for studying behavior change*. New York: Pergamon Press.
- Bernstein, G. A., Borchardt, C. M., Perwien, A. R., Crosby, R. D., Kushner, M. G., Thuras, P. D., & Last, C. G. (2000). Imipramine plus cognitive-behavioral therapy in the treatment of school refusal. *Journal of the American Academy of Child and Adolescent Psychiatry*, 39, 276–283.
- Cain, C. K., Blouin, A. M., & Barad, M. (2003). Temporally massed CS presentations generate more fear extinction than spaced presentations. *Journal of Experimental Psychology: Animal Behavior Processes*, 29, 323–333.
- Craske, M. G., Kircanski, K., Zelikowsky, M., Mystkowski, J., Chowdhury, N., & Baker, A. (2008). Optimizing inhibitory learning during exposure therapy. *Behaviour Research and Therapy*, 46, 5–27.
- Foa, E. B., Jameson, J. S., Turner, R. M., & Payne, L. L. (1980). Massed vs. spaced exposure sessions in the treatment of agoraphobia. *Behaviour Research and Therapy*, 18, 333–338.
- Granell de Aldaz, E., Vivas, E., Gelfand, D. M., & Feldman, L. (1984). Estimating the prevalence of school refusal and school-related fears. A Venezuelan sample. *Journal of Nervous and Mental Disease*, 172, 722–729.
- Guy, W. (1976). *Assessment manual for psychopharmacology*. Washington, DC: U.S. Government Printing Office.
- Hibbett, A., & Fogelman, K. (1990). Future lives of truants: Family formation and health-related behaviour. *Journal of Educational Psychology*, 60, 171–179.
- Higa, C. K., Daleiden, E. L., & Chorpita, B. F. (2002). Psychometric properties and clinical utility of the School Refusal Assessment Scale in a multiethnic sample. *Journal of Psychopathology and Behavioral Assessment*, 24, 247–258.
- Kearney, C. A. (1993). Depression and school refusal behavior: A review with comments on classification and treatment. *Journal of School Psychology*, 31, 267–279.
- Kearney, C. A. (1995). School refusal behavior. In A. R. Eisen, C. A. Kearney, & C. E. Schaefer (Eds.), *Clinical handbook of anxiety disorders in children and adolescents* (pp. 19–52). Northvale, NJ: Jason Aronson.
- Kearney, C. A. (2001). *School refusal behavior in youth: A functional approach to assessment and treatment*. Washington, DC: American Psychological Association.
- Kearney, C. A. (2002). Identifying the function of school refusal behavior: A revision of the School Refusal Assessment Scale. *Journal of Psychopathology and Behavioral Assessment*, 24, 235–245.
- Kearney, C. A., & Albano, A. M. (2000). *When children refuse school: A cognitive-behavioral therapy approach (Therapist guide)*. San Antonio, TX: Psychological Corporation.
- Kearney, C. A., & Bates, M. (2005). Addressing school refusal behavior: Suggestions for frontline professionals. *Children and Schools*, 27, 207–216.
- Kearney, C. A., Pursell, C., & Alvarez, K. (2001). Treatment of school refusal in children with mixed functional profiles. *Cognitive and Behavioral Practice*, 8, 3–11.
- Kearney, C. A., & Silverman, W. K. (1990). A preliminary analysis of a functional model of assessment and treatment for school refusal behavior. *Behavior Modification*, 14, 340–366.
- Kearney, C. A., & Silverman, W. K. (1993). Measuring the function of school refusal behavior: The School Refusal Assessment Scale. *Journal of Clinical Child Psychology*, 22, 85–95.
- Kearney, C. A., & Silverman, W. K. (1999). Functionally-based prescriptive and nonprescriptive treatment for children and adolescents with school refusal behavior. *Behavior Therapy*, 30, 673–695.
- Kearney, C. A., & Tillotson, C. A. (1998). School attendance. In T. S. Watson, & F. M. Gresham (Eds.), *Handbook of child behavior therapy* (pp. 143–161). New York: Plenum Press.
- Kendall, P. C., Chu, B., Gifford, A., Hayes, C., & Nauta, M. (1998). Breathing life into a manual: Flexibility and creativity with manual-based treatments. *Cognitive and Behavioral Practice*, 5, 177–198.
- Kendall, P. C., & Hedtke, K. A. (2006). *Cognitive-behavioral therapy for anxious children (Therapist manual)*, (3rd ed.) Ardmore, PA: Workbook Publishing.
- King, N. J., Heyne, D., & Ollendick, T. H. (2005). Cognitive-behavioral treatments for anxiety and phobic disorders in children and adolescents: A review. *Behavioral Disorders*, 30, 241–257.
- King, N. J., Tonge, B. J., Heyne, D., Pritchard, M., Rollings, S., Young, D., Myerson, N., & Ollendick, T. H. (1998). Cognitive-behavioral treatment of school-refusing children: A controlled evaluation. *Journal of the American Academy of Child and Adolescent Psychiatry*, 37, 395–403.
- King, N. J., Tonge, B. J., Heyne, D., Turner, S., Pritchard, M., Young, D., Rollings, S., Myerson, N., & Ollendick, T. H. (2001). Cognitive-

- behavioural treatment of school-refusing children: Maintenance of improvement at 3- to 5-year follow-up. *Scandinavian Journal of Behaviour Therapy*, 30, 85–89.
- Kovacs, M. (1985). The Children's Depression Inventory (CDI). *Psychopharmacology Bulletin*, 21, 995–998.
- Last, C. G., Hansen, C., & Franco, N. (1998). Cognitive-behavioral treatment of school phobia. *Journal of the American Academy of Child and Adolescent Psychiatry*, 37, 404–411.
- Layne, A. E., Bernstein, G. A., Egan, E. A., & Kushner, M. G. (2003). Predictors of treatment response in anxious-depressed adolescents with school refusal. *Journal of the American Academy of Child and Adolescent Psychiatry*, 42, 319–326.
- Leon, A. C., Shear, M. K., Portera, L., & Klerman, G. L. (1992). Assessing impairment in patients with panic disorder: The Sheehan Disability Scale. *Social Psychiatry and Psychiatric Epidemiology*, 27, 78–82.
- March, J. S., Parker, J. D., Sullivan, K., Stallings, P., & Conners, C. K. (1997). The Multidimensional Anxiety Scale for Children (MASC): Factor structure, reliability, and validity. *Journal of the American Academy of Child and Adolescent Psychiatry*, 36, 554–565.
- Moffitt, C. E., Chorpita, B. F., & Fernandez, S. N. (2003). Intensive cognitive-behavioral treatment of school refusal behavior. *Cognitive and Behavioral Practice*, 10, 51–60.
- Ollendick, T. H., & Mayer, J. A. (1984). School phobia. In S. M. Turner (Ed.), *Behavioral theories and treatment of anxiety* (New York: Plenum Press).
- Roblek, T., & Piacentini, J. (2005). Cognitive-behavior therapy for childhood anxiety disorders. *Child and Adolescent Psychiatric Clinics of North America*, 14, 863–876 x.
- Tolin, D. F., & Franklin, M. E. (2002). Prospects for the use of cognitive-behavioral therapy in childhood obsessive-compulsive disorder. *Expert Review of Neurotherapeutics*, 2, 89–98.
- Whiting, S., & Tolin, D. F. (2008, November). Psychosocial impairment in childhood anxiety: Utility and validation of the Sheehan Disability Scale-Child/Parent Version. Paper presented at the Annual Meeting of the Association of Behavioral and Cognitive Therapies, Orlando, FL.
- Young, L. C. (1941). On randomness in ordered sequences. *Annals of Mathematical Statistics*, 12, 153–162.

Address correspondence to David F. Tolin, Ph.D., Anxiety Disorders Center, The Institute of Living, Hartford Hospital, 200 Retreat Avenue, Hartford, CT 06106; e-mail: dtolin@harthosp.org.

Received: December 4, 2008

Accepted: February 16, 2009

Available online 17 June 2009